

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Some Applications of Trigonometry

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

From a point 10 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/10$. Therefore height = $10/\sqrt{3}$ m.

Question 2

From a point 20 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/20$. Therefore height = 20 m.

Question 3

From a point 15 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/15$. Therefore height = $15\sqrt{3}$ m.

Question 4

From a point 12 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/12$. Therefore height = 12 m.

Question 5

From a point 8 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/8$. Therefore height = $8/\sqrt{3}$ m.

Question 6

From a point 25 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/25$. Therefore height = 25 m.

Question 7

From a point 6 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/6$. Therefore height = $6\sqrt{3}$ m.

Some Applications of Trigonometry - Continued

Question 8

From a point 18 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/18$. Therefore height = $18/\sqrt{3}$ m.

Question 9

From a point 30 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/30$. Therefore height = 30 m.

Question 10

From a point 14 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/14$. Therefore height = $14\sqrt{3}$ m.

Question 11

From a point 16 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/16$. Therefore height = 16 m.

Question 12

From a point 20 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/20$. Therefore height = $20/\sqrt{3}$ m.

Question 13

From a point 9 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/9$. Therefore height = $9\sqrt{3}$ m.

Question 14

From a point 24 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/24$. Therefore height = 24 m.

Question 15

From a point 10 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/10$. Therefore height = $10\sqrt{3}$ m.

Some Applications of Trigonometry - Continued

Question 16

From a point 12 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/12$. Therefore height = $12/\sqrt{3}$ m.

Question 17

From a point 35 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/35$. Therefore height = 35 m.

Question 18

From a point 5 m from a tower, the angle of elevation of the top is 60 degrees. Find its height, ignoring eye level.

Answer:

$\tan 60^\circ = \text{height}/5$. Therefore height = $5\sqrt{3}$ m.

Question 19

From a point 40 m from a tower, the angle of elevation of the top is 30 degrees. Find its height, ignoring eye level.

Answer:

$\tan 30^\circ = \text{height}/40$. Therefore height = $40/\sqrt{3}$ m.

Question 20

From a point 7 m from a tower, the angle of elevation of the top is 45 degrees. Find its height, ignoring eye level.

Answer:

$\tan 45^\circ = \text{height}/7$. Therefore height = 7 m.

Question 21

State the main formula, rule or theorem used in Some Applications of Trigonometry.

Answer:

$\tan \theta = \text{perpendicular}/\text{base}$ is most often used for height-and-distance questions.

Question 22

What do the important symbols or quantities in Some Applications of Trigonometry represent?

Answer:

Angle of elevation is measured upward from the horizontal; angle of depression is measured downward.

Question 23

Write the correct method for solving a Some Applications of Trigonometry numerical.

Answer:

Draw a labelled right triangle, identify the known side, choose the correct trigonometric ratio and state units.

Some Applications of Trigonometry - Continued

Question 24

Which unit, sign convention or condition must be checked in a Some Applications of Trigonometry answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Some Applications of Trigonometry be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Some Applications of Trigonometry questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Some Applications of Trigonometry numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Some Applications of Trigonometry?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Question 29

How can a student decide which formula to use in Some Applications of Trigonometry?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Some Applications of Trigonometry?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: CBSE Mathematics curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.